QUANTUM OPTICS 2025

Hypothesis:

In the 1960s, Lamb and Scully showed that the photoelectric effect could also be explained by a semiclassical theory, in addition to Einstein's quantized theory. In 1986, Grangier, Roger, and Aspect (GRA) designed an experiment using a 50/50 beam splitter to determine between the two possibilities. *Using the ThorLab Quantum Optics kit, an experiment to determine the nature of light will be designed, assembled, conducted, documented, analyzed with uncertainty quantified, and interpreted to present quantitative measurements that will be compared against the predictions of the two competing theories.*

History:

In 1905, Albert Einstein wrote the paper that would win him a Nobel prize. Building on the work of Max Planck, Einstein hoped to explain the photoelectric effect by proposing that light is quantized. Where Planck used quantization as a simple mathematical trick, Einstein asserted its physical reality. The photoelectric effect is generally considered to be the first absolute proof of quantization, the first theory that required it on a fundamental level. However, this is not the case. In the 1960s, Lamb and Scully showed that the photoelectric effect can be explained by a semiclassical theory in which atomic energy levels are quantized but light itself is not. Because of this new theory, some questioned whether light is quantized at all. In 1986, Grangier, Roger, and Aspect (GRA) devised an experiment that could prove with certainty that light is quantized. They realized that the predicted behavior of light in a 50/50 beam splitter is different in the semiclassical theory and the quantum theory. Very carefully-prepared light was put through such a beam splitter so that its outputs could be observed.

Background:

In the quantum theory, there is a minimum irradiance that can be put into the beamsplitter: one photon at a time. The photon cannot be split further, and as such it can only go to one output. Therefore, we simply need to observe what happens when a beam splitter is put in the path of a beam of single photons (the semiclassical theory would just call it very weak light). If there are two simultaneous outputs, we say the outputs are correlated, and the quantum theory is disproved. If light only comes out of one output at a time, then the outputs are said to be anticorrelated, and the semiclassical theory is disproved.

A challenge is that most light sources, even single-photon emitters, will not always produce a single photon. Quantum theory predicts several states of light, known as **Fock states**. Fock states contain a well-defined number of photons, n. Single photons are described by the Fock state $|1\rangle$. Most light sources, even single-photon emitters, do not produce a perfect $|1\rangle$ state but instead emit a mixture of states. For example, an imperfect single-photon source might sometimes emit $|2\rangle$ or even $|0\rangle$, making it difficult to confirm that only one photon is present per pulse. Regular light sources emit coherent states, which are superpositions of Fock states. As such, coherent states do not have a well-defined photon number. No matter how attenuated one makes a coherent source, it will never output single photons.

Generating Fock states is difficult. For their paper, Grangier, Roger, and Aspect used a finicky and inefficient atomic calcium cascade. Our experiment uses spontaneous parametric down-conversion, a simpler method to implement. A proper theoretical description requires the

machinery of nonlinear quantum optics, so the description here is necessarily simplified and qualitative. More mathematical detail can be found in the identified reading.

Spontaneous Parametric Down Conversion:

Light is an electromagnetic phenomenon; a light wave contains an electric field and a magnetic field. When the electric field of a light beam is incident on matter, it causes the atoms and molecules in that matter to become polarized. In most systems, the resulting polarization density is very well-approximated by a linear relationship with the electric field strength. However, when very intense light is applied, the electric field is so strong that the polarization density begins to behave nonlinearly. In most materials, this requires impractically powerful light, but certain crystals exhibit useful nonlinear behavior at attainable field strengths. For this experiment, we will use a nonlinear beta-barium borate (BBO) crystal to achieve spontaneous parametric down conversion (SPDC).

SPDC is a nonlinear optical process in which a photon with energy E_1 enters a nonlinear crystal and gets converted into two output photons with energies E_2 and E_3 such that $E_1=E_2+E_3$. SPDC has many applications in experimental optics, as the photons it generates are entangled. Here, though, it is employed because it can generate Fock states of light from incident coherent states. In other words, SPDC allows for generation of single photons.

Strictly speaking, SPDC generates photon pairs, but conservation of momentum dictates that the two photons move in different directions, so it's easy to position our beam splitter in such a way that only one photon enters it at a time. Rather than discard the other photon, we can position a detector in its path and use it as a herald. Detecting the herald photon tells us that the other photon of the pair should be just about to be detected (it is slowed slightly by its passage through the beamsplitter). We can discard any beamsplitter photon detections that aren't preceded by a herald detection, allowing for a significant reduction in the error introduced by background light.

This pair of correlated single photons is the essence of this experiment. A strong laser will generate intense coherent light which is then pointed into our BBO crystal. The nonlinear process of SPDC will generate photon pairs in a Fock state, allowing us to pass individual photons through a 50/50 beamsplitter with a detector at each output. By observing whether the outputs are correlated, we can determine whether or not light is really quantized.

The Second-Order Correlation Function:

How will we actually quantify the results of our experiment? How do we use the raw data from the detector to measure correlation? These questions are answered with the second-order correlation function $g^{(2)}(\tau)$. This function takes as input the time-averaged irradiance of each output beam from the beamsplitter and produces a dimensionless number that quantifies the degree to which the count rates are correlated. It has the form

$$g^{(2)}(\tau) = \frac{\langle I_A(t+\tau) \cdot I_B(t) \rangle}{\langle I_A(t+\tau) \rangle \cdot \langle I_B(t) \rangle},$$

where I_A and I_B are the irradiances at output detectors A and B and the brackets $\langle\rangle$ denote a time average. We care about the correlation between intensities at the same time, which is found by taking $\tau=0$:

$$g^{(2)}(0) = \frac{\langle I_A \cdot I_B \rangle}{\langle I_A \rangle \cdot \langle I_B \rangle}.$$

However, our detectors measure the number of photons detected per unit time instead of intensity. To get $g^{(2)}$ in terms of the count rates, we need to introduce the time window Δt. In order to measure a correlation, a photon must arrive at each detector within the interval Δt. Because of our heralded photon scheme, these detections will only be counted if a photon was also detected by the herald detector within that same time interval. We then get a $g^{(2)}(0)$ for the GRA experiment of the form

$$g^{(2)}_{GRA} = \frac{R_{TAB} \cdot R_T}{R_{TA} \cdot R_{TB}},$$

where R_T is the count rate of the herald detector (also called the trigger detector), R_{TA} is the rate of coincidence counts between detectors T and A, R_{TB} the coincidence count rate between T and B, and R_{TAB} the coincidence rate between all three.

A second-order correlation function equal to 1 indicates that the count rates R_A"and" R_B are completely uncorrelated. If the function is greater than one, then the rates are correlated. The higher the value, the more correlated they are, so the more likely a detection at A is followed by a detection at B. If the function is less than one, then the rates are anticorrelated, so it is unlikely to see a detection at A alongside a detection at B.

The semiclassical theory predicts that $g^{(2)} \geq 1$, because there is always a field at each output so there will never be anticorrelation. However, for single photons, the quantum theory predicts $g^{(2)} < 1$, because the photon must go one way or the other, so only one detector will be activated by each photon.

In the course of the experiment, count rates and coincidence count rates will be recorded for each detector. The second-order correlation function can then be calculated, and the resulting value used to decide whether or not light is quantized.

Objectives:

1. Become familiar with how to use and potentially align several common optical components and understand their purpose. (If the setup is well-aligned, I'm not going to make you change it. However, realigning optics is a MAJOR part of optics research and there is a good chance things could get unaligned, so be ready for that possibility.)
2. Understand how the setup creates a source of single photons using spontaneous parametric down conversion with a BBO crystal. Use this source along with a 50/50 beamsplitter and three single-photon detectors to recreate the Grangier-Roger-Aspect experiment.
3. Take measurements of the photon count rate at each detector and the coincidence count rate between each detector using the ThorLabs EDU-QOPT1 software, the Swabian Instruments Time Tagger software, or the Swabian Instruments Time Tagger API. Whichever method is chosen, error analysis must be performed as well in order to propagate your error of the second order correlation function.
4. Compare your experimental results against the predictions of the two competing theories and determine whether the quantum or semiclassical approach is supported.
5. Using what you have learned, explain the counterintuitive results of the HBT experiment with the pair source.

Equipment:
The equipment consists of parts that came with the ThorLabs Quantum Optics Kit EDU-QOPT1, with the exception of the computer. The complete list of parts and part numbers can be found in chapter 5 of the kit's manual (page 38). Especially important components include:

- 405 nm, 20 mW laser diode L405P20 and mount LDM9T, referred to as the "pump laser"
- "K-Cube" laser diode controller KLD101, for controlling the pump laser
- 635 nm, 0.9 mW USB diode laser PL202, referred to as the "alignment laser"
- Mounted beta-barium borate (BBO) crystal NLCQ3 with 1 inch diameter and 3 mm thickness
- Half-inch diameter axicon with 3° opening angle for 636 nm light (no part number), with lens tube SM05L10 for mounting
- 515 nm long-pass colored glass filer FGL515, 25mm diameter
- Mounted, nonpolarizing, anti-reflection coated beamsplitter CCM5-BS017 for wavelengths 700-1100 nm
- Three single photon detectors SPDMA with 500 μm detector area, optimized for 350-1100 nm
- Swabian Instruments Time Tagger 20
- Beam stop
- Two laser safety glasses LG3
- A computer with the ThorLabs EDU-QOPT1 software installed

Turning on the laser:
More details can be found in the ThorLab Quantum Optics kit manual (p. 66), but to summarize:

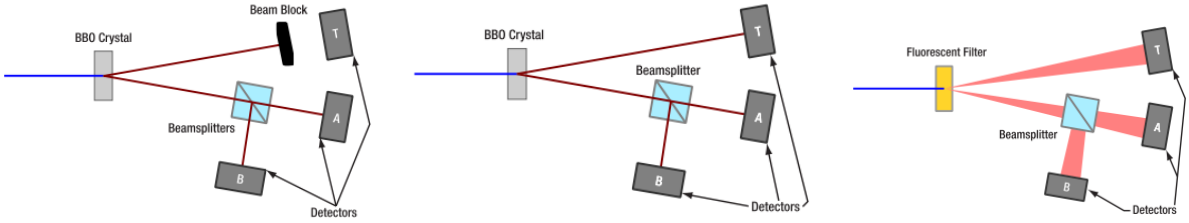
1. Turn on the laser diode switch and make sure still set to 25°C.
2. Turn on the power on the laser diode controller.
3. Use the wheel to turn up the current to 15 mA.
4. **Put on your safely glasses**. Enable the pump. If you put a piece of paper in the beam path, with your glasses on you will see a green spot.
5. You will now use the wheel to carefully turn up the power; feel free to use the paper to monitor what happens to the green spot as you do so. If you are not using the BBO crystal, 26 mA is a good value at which to operate. The BBO crystal requires lasing, which happens at a jump up in the intensity of that green spot around 34 mA. Stay near this value if using the BBO crystal and about 8 mA below that lasing threshold otherwise.

Procedure:

1. In the ThorLab Quantum Optics manual, look over chapter 3 (particularly pages 4-5 and sections 3.2-3.5 and 3.9) to understand the second-order correlation function $g^{(2)}(\tau)$, its interpretation, and its form for the Grangier-Roger-Aspect experiment specifically. Deriving the function requires familiarity with the mathematics of quantum mechanics and quantum optics, so it is not required. Also read sections 7.3-7.5. In addition to the copy provided in the lab, it can be found online at: https://www.thorlabs.com/thorproduct.cfm?partnumber=EDU-QOP1
2. Write down the name and purpose of the following optical components: detectors, BBO crystal, beam splitter and fluorescent filter. Draw a diagram of the detector and look up how you would try to align it if the alignment were off. There are other things you could need to align too, but the detectors would be a good place to attempt if the alignment

appears decent. This graded as part of the pre-lab background, which should be longer than other labs as there are not many formulas to discuss for this lab. See pages 48-64.

3. You should also make a diagram of the optics table similar to the one on page 45. Add the beam path to the figure. The figure needs simplified, though. Your audience does not need the part numbers. Some parts may also not be necessary. Unless you do a full realignment of the setup (meaning more than just tweaking the detectors), you will not use the axicon. If you do not use something, it should not be on your figure.
4. In the pre-lab, discuss the predictions of both the semiclassical and quantum theories for light incident on a 50/50 beam splitter.
5. Alignment of the optics setup may or may not be necessary. It was setup, but sometimes things get bumped or even thermal fluctuations can affect things. Using a piece of paper, see if the beam appears to be going into the center of the optics up to the BBO crystal, after which you can no longer see the beam. If the beam appears centered, section 7.5.4 (page 87) is a test measurement (using the same setup as in 9.4) used to verify that the GRA experiment is setup and aligned well. If it is, please continue with the procedure.



Experiments 9.3, 9.4 and 9.5, respectively

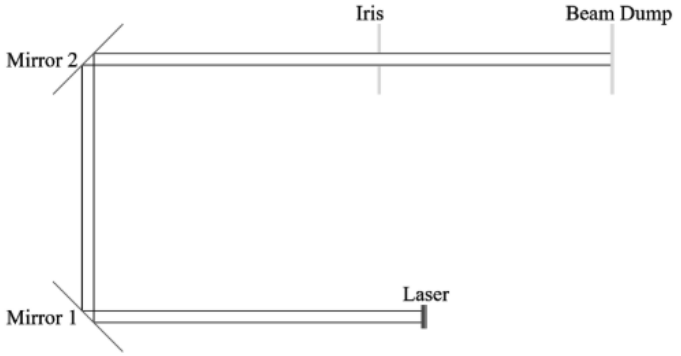
6. Do experiments 9.3 (HBT tab), 9.4 and 9.5 (GRA tab) in any order. Two involve a BBO crystal, where you'll want the laser at 34 mA. For 9.5, you'll want the laser at 26 mA. For 9.3, you only need to add a beam block to detector T, compared to 9.4.
7. Record data for at least 10 tries per experiment and calculate uncertainty. Make sure you can find the uncertainty for all rates, as you will need those to correctly propagate error.
8. Calculate the value and error of $g^{(2)}(0)$ and then interpret the result to decide which theory is supported. Explain the counterintuitive results in the HBT experiment.
9. When finished, make sure you turn off the detectors, laser diode controller and pump laser.

Guidance if things get misaligned

1. Don't panic. While there are extra steps you will have to go through, you will learn a lot about optics by doing this extra work. This practice is seen as very valuable learning by the optics professors here. Some like to have their students go through these experiments to learn how to do things in their research labs.
2. Tip 1: It is often helpful to refer to optical elements by the order in which the light hits them. We use the terms "**upstream**" and "**downstream**" as if the light beam is a river. An upstream optic gets hit by the light before a downstream optic.
3. Tip 2: You should coarsely align the equipment by placing it at a close enough position, clamping it down, and then using the knobs to align it. If alignment is necessary, irises places as far away as possible on the optical table in a straight are very handy for aligning the beam; the farther away, the better leveled your beam path will be.
4. Tip 3: You may also find it easier to align with an index card or piece of paper rather than the white beam stop.

5. Assess how bad the alignment is. If the beam goes through the center of the optics up to the BBO crystal, you might be able to get by with minor adjustments. You might only need to align a detector or two (see section 7.4.3 on detector positioning). You might be able to figure out which detector(s) seem off by comparing with the sample data counts.
6. If you decide you need to realign from earlier in the setup, follow and perform section 7.4.2 in the ThorLabs manual: Aligning Both Pump and Alignment Laser on Same Path. (Page 75)
7. Set up the axicon, BBO crystal, beamsplitter, and detectors using sections 7.4.3-7.4.5 in the Thorlabs manual. Setting up the detectors may be hard with one person since the table is high and the detectors point inwards, so this will most likely need two people to be able to align the detectors.
8. Before starting section 7.5, do experiment 9.2 (Photon Pair Source) and download the data from the GRA tab. You should also report on this in your lab notebook and written lab report if you go through this alignment.

Optional: Student written summary of guidance in the alignment section of the manual



Above figure: 2 mirror and iris experiment:

1. Close the iris in the figure above. Place the iris upstream of the beam dump according to the manual 8 holes from the short edge of the breadboard (The iris and beam dump must be facing in such a way that you can imagine a line starting from the center of the beam dump, going through the iris, and then to the laser)
2. Place the first mirror downstream of the red laser on the 6th hole from the short edge of the breadboard and align the laser on the center of the mirror
3. Place the second mirror (magnetic mounted for the red laser) downstream of the first mirror 6 holes from the short edge of the breadboard so the light can go through the iris and into the beam dump
4. Rotate the post of the first mirror so the laser hits the center of the second mirror (coarse alignment)
5. Rotate the post of the second mirror so the laser hits the iris close to the aperture (coarse alignment)
6. Use the adjustment screws to make the light from the first mirror hit the center of the second mirror and make the light from the second mirror hit the beam dump (fine alignment)
   a. Close the iris and then adjust the first mirror so the laser hits the iris aperture
   b. Open the iris and then adjust the second mirror so the laser hits the center of the beam dump (this will be easier if the iris on the beam dump is closed and the room lights are on)
   c. Repeat (a) and (b) until the laser goes through the center of the iris aperture and hits the center of the beam dump (this process is called walking the beam)

7. Place the half-wave plate 16 holes from the long edge and 3 holes from the short edge of the breadboard with the dial, set at 45°, facing upstream of the blue laser (There is a gauge on the bottom of the rotation mount that you can align the number to)
8. Repeat steps 2-6 with the blue laser set to 30 mA and 2 additional mirrors and only move the magnetic mirror off of the magnetic mount (you should be able to make both lasers go through the iris and hit the center of the beam dump without adjusting anything other than replacing the magnetic mounted mirror)